

Celebal Technologies Internship – Capstone Project

CityReads Data Engineering Pipeline

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Internship: Celebal Technologies Data Engineering Internship

Project: CityReads Data Engineering Capstone

Objective

The objective of this capstone project is to design and implement an end-to-end Data Engineering pipeline for the CityReads dataset. The project demonstrates incremental data ingestion, watermark-based processing, data cleaning, validation, rejected-record handling, business KPI generation, and pipeline health auditing using the Medallion Architecture.

Dataset Description

Five datasets were used in this project.

books.csv

The books dataset contains information about books available in the CityReads library.

customers.csv

The customers dataset contains information about CityReads customers and members.

orders.csv

The orders dataset contains customer book-order and transaction information.

loans.csv

The loans dataset contains information about book loans, due dates, and return dates.

reviews.csv

The reviews dataset contains customer ratings and review information for books.

Initial dataset record counts:

- Books – 190
- Customers – 2,800
- Orders – 28,336
- Loans – 9,614
- Reviews – 5,621

Total source records: 46,561

Software Requirements

- Python
- Pandas
- MySQL
- MySQL Workbench
- SQL
- Visual Studio Code
- Git
- GitHub

Q1. Prepare Source Dataset and MySQL Database

Explanation

The five CityReads source datasets were prepared and organized inside the project directory. A MySQL database named **cityreads** was created for structured data storage and SQL analysis.

The following source tables were created:

- books
- customers
- orders
- loans
- reviews

The source datasets were then loaded into MySQL and validated using SQL COUNT queries.

Screenshot

```
Directory: C:\Users\HP\OneDrive\Desktop\Capstone Project\data\cityreads_dataset

Mode                LastWriteTime         Length Name
----                -
-a----            8/12/2026   9:37 AM           12963 books.csv
-a----            8/12/2026   9:37 AM          193137 customers.csv
-a----            8/12/2026   9:37 AM          426135 loans.csv
-a----            8/12/2026   9:37 AM         1022832 orders.csv
-a----            8/12/2026   9:37 AM          482863 reviews.csv

PS C:\Users\HP\OneDrive\Desktop\Capstone Project> python -c "import pandas as pd; p='data/cityreads_dataset'; print('books:',len(pd.read_csv(p+'/books.csv'))); print('customers:',len(pd.read_csv(p+'/customers.csv'))); print('loans:',len(pd.read_csv(p+'/loans.csv'))); print('orders:',len(pd.read_csv(p+'/orders.csv'))); print('reviews:',len(pd.read_csv(p+'/reviews.csv')))"
books: 190
customers: 2800
loans: 9614
orders: 28336
reviews: 5621
PS C:\Users\HP\OneDrive\Desktop\Capstone Project>
```

Q2. Validate Source Dataset

Explanation

The source datasets were validated after loading them into MySQL.

The final source record counts were:

- Books – 190
- Customers – 2,800
- Orders – 28,336
- Loans – 9,614
- Reviews – 5,621

Total source records: 46,561

Screenshot

The screenshot displays a SQL query in a text editor, followed by its execution results in a 'Result Grid'.

SQL Query:

```
138 • SELECT 'books' AS table_name, COUNT(*) AS row_count FROM books
139 UNION ALL
140 SELECT 'customers', COUNT(*) FROM customers
141 UNION ALL
142 SELECT 'orders', COUNT(*) FROM orders
143 UNION ALL
144 SELECT 'loans', COUNT(*) FROM loans
145 UNION ALL
146 SELECT 'reviews', COUNT(*) FROM reviews;
147
148
```

Result Grid:

table_name	row_count
books	190
customers	2800
orders	28336
loans	9614
reviews	5621

Q3. Implement Bronze Layer Incremental Ingestion

Explanation

The Bronze layer was implemented using Python to perform incremental data ingestion.

The ingestion process reads the source CSV files, checks the previous watermark, identifies new records, writes the records into Bronze files, and updates the watermark state.

Separate Bronze directories were created for:

- Books
- Customers
- Orders
- Loans
- Reviews

Screenshot

```
=====
CITYREADS - BRONZE LAYER INGESTION
=====
```

```
Enter MySQL root password:
```

```
MySQL connection successful.
```

```
Batch ID: BATCH_20260813_185424
```

```
-----
TABLE: BOOKS
-----
```

```
Source rows      : 190
Previous watermark : 2000-01-01 00:00:00
New rows         : 190
Bronze rows loaded : 190
New watermark     : 2023-09-07 00:00:00
Bronze table      : bronze_books
```

```
-----
TABLE: CUSTOMERS
-----
```

```
Source rows      : 2800
Previous watermark : 2024-10-01 00:00:00
New rows         : 0
No new records found.
```

```
-----
TABLE: ORDERS
-----
```

```
Source rows      : 28336
Previous watermark : 2024-12-31 00:00:00
New rows         : 0
No new records found.
```

```
=====
CITYREADS - BRONZE LAYER INGESTION
=====
```

```
Enter MySQL root password:
```

```
MySQL connection successful.
```

```
Batch ID: BATCH_20260813_185812
```

```
-----
TABLE: BOOKS
-----
```

```
Source rows      : 190
```

```
Previous watermark : 2023-09-07 00:00:00
```

```
New rows         : 0
```

```
No new records found.
```

```
-----
TABLE: CUSTOMERS
-----
```

```
Source rows      : 2800
```

```
Previous watermark : 2024-10-01 00:00:00
```

```
New rows         : 0
```

```
No new records found.
```

```
-----
TABLE: ORDERS
-----
```

```
Source rows      : 28336
```

```
Previous watermark : 2024-12-31 00:00:00
```

```
New rows         : 0
```

```
No new records found.
```

```
-----
TABLE: LOANS
-----
```

```
Source rows      : 9614
```

```
Previous watermark : 2024-12-26 00:00:00
```

```
New rows         : 0
```

```
No new records found.
```

TABLE: REVIEWS

Source rows : 5621
Previous watermark : 2024-12-31 23:53:34
New rows : 0
No new records found.

=====

BRONZE INGESTION SUMMARY

=====

Total new rows : 0

Pipeline metadata:

bronze_books : 2023-09-07 00:00:00 | rows=190 | status=SUCCESS
bronze_customers : 2024-10-01 00:00:00 | rows=2800 | status=SUCCESS
bronze_loans : 2024-12-26 00:00:00 | rows=9614 | status=SUCCESS
bronze_orders : 2024-12-31 00:00:00 | rows=28336 | status=SUCCESS
bronze_reviews : 2024-12-31 23:53:34 | rows=5621 | status=SUCCESS

MySQL connection closed.

Bronze ingestion completed.

Result Grid   Filter Rows: Edit:    Export				
	table_name	last_loaded_at	rows_loaded	status
▶	bronze_books	2023-09-07 00:00:00	190	SUCCESS
	bronze_customers	2024-10-01 00:00:00	2800	SUCCESS
	bronze_loans	2024-12-26 00:00:00	9614	SUCCESS
	bronze_orders	2024-12-31 00:00:00	28336	SUCCESS
	bronze_reviews	2024-12-31 23:53:34	5621	SUCCESS
✱	NULL	NULL	NULL	NULL

Q4. Implement Watermark-Based Processing

Explanation

A **watermark.json** file was created to maintain the latest processed record position for every dataset.

The final watermark values were:

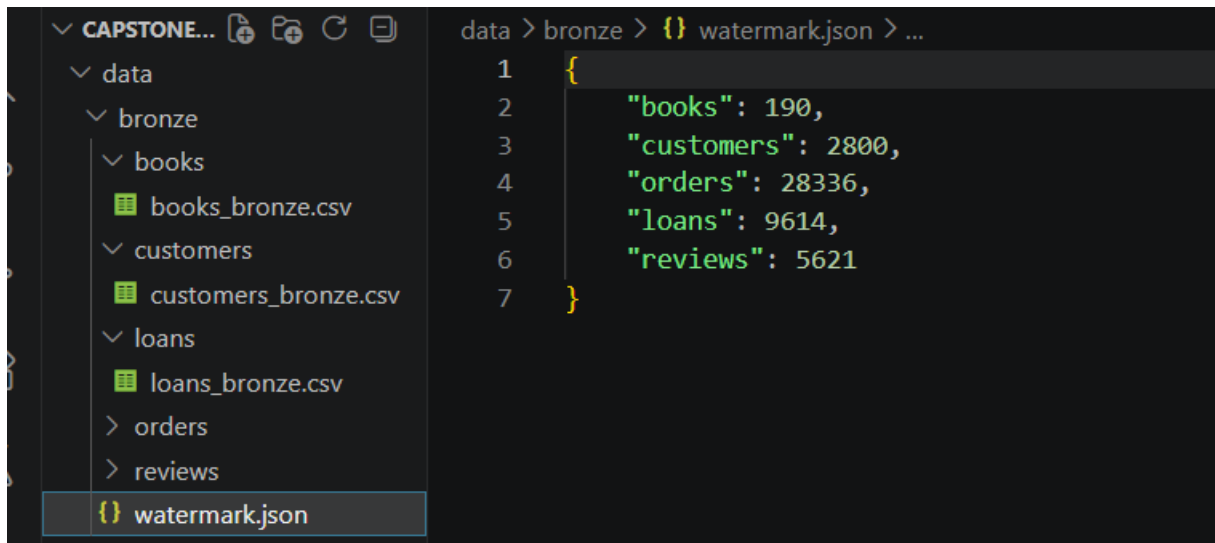
- Books – 190
- Customers – 2,800
- Orders – 28,336
- Loans – 9,614
- Reviews – 5,621

The ingestion script was executed again to verify incremental processing.

Total New Rows: 0

This confirms that previously processed records were not ingested again.

Screenshot



Q5. Implement Silver Layer Transformation

Explanation

The Silver layer was implemented to clean, validate, and transform the Bronze data.

The transformation process performs:

- Data validation
- Duplicate detection
- Data cleaning
- Business-rule validation
- Invalid-record detection
- Rejected-record handling

The transformation produced the following results:

Dataset	Bronze Rows	Silver Rows
----------------	--------------------	--------------------

Books	190	190
-------	-----	-----

Customers	2,800	2,766
-----------	-------	-------

Orders	28,336	27,354
--------	--------	--------

Loans	9,614	9,402
-------	-------	-------

Reviews	5,621	5,454
---------	-------	-------

Total Bronze Rows: 46,561

Total Silver Rows: 45,166

Total Rejected Rows: 1395

Screenshot

CITYREADS - SILVER LAYER TRANSFORMATION

BOOKS

Bronze rows: 190
Duplicates removed: 0
Silver books: 190

CUSTOMERS

Bronze rows: 2800
Duplicates removed: 0
Silver customers: 2766

ORDERS

Bronze rows: 28336
Duplicates removed: 0
Silver orders: 27354

LOANS

Bronze rows: 9614
Duplicates removed: 0
Silver loans: 9402

REVIEWS

Bronze rows: 5621
Duplicates removed: 0
Silver reviews: 5454

SILVER TRANSFORMATION COMPLETED

Total rejected rows : 1395
Rejected file : c:\Users\HP\OneDrive\Desktop\Capstone Project\data\silver\rejected\silver_rejected_rows.csv

Silver output:

Books : 190
Customers : 2766
Orders : 27354
Loans : 9402
Reviews : 5454

Q6. Handle Rejected Records

Explanation

Records that failed the Silver-layer validation rules were stored separately instead of being discarded.

The rejected records were stored in:

data/silver/rejected/silver_rejected_rows.csv

A total of **1395 records** were rejected.

The rejected records contain information about the source record and the reason for rejection, providing traceability for data-quality issues.

Screenshot

REJECTION SUMMARY

table_name	rejection_reason	count
customers	INVALID_MEMBERSHIP	12
customers	NULL_EMAIL	22
loans	INVALID_CUSTOMER_FK	136
loans	INVALID_DUE_DATE	76
orders	INVALID_BOOK_FK	161
orders	INVALID_CUSTOMER_FK	320
orders	INVALID_QUANTITY	145
orders	INVALID_STATUS	356
reviews	INVALID_CUSTOMER_FK	68
reviews	INVALID_RATING	99

TOTAL: 1395

PS C:\Users\HP\OneDrive\Desktop\Capstone Project> █

```
=====
CITYREADS - SILVER LAYER TRANSFORMATION
=====
```

```
BOOKS
-----
```

```
Bronze rows: 190
Duplicates removed: 0
Silver books: 190
```

```
CUSTOMERS
-----
```

```
Bronze rows: 2800
Duplicates removed: 0
Silver customers: 2766
```

```
ORDERS
-----
```

```
Bronze rows: 28336
Duplicates removed: 0
Silver orders: 27354
```

```
LOANS
-----
```

```
Bronze rows: 9614
Duplicates removed: 0
Silver loans: 9402
```

```
REVIEWS
-----
```

```
Bronze rows: 5621
Duplicates removed: 0
Silver reviews: 5454
```

```
=====
SILVER TRANSFORMATION COMPLETED
=====
```

```
Total rejected rows : 1395
Rejected file       : C:\Users\HP\OneDrive\Desktop\Capstone Project\data\silver\rejected\silver
_rejected_rows.csv
```

Silver output:

Books : 190
Customers : 2766
Orders : 27354
Loans : 9402
Reviews : 5454

=====

=====

SYNCING SILVER LAYER TO MYSQL

=====

Enter MySQL root password:
MySQL connection successful.

TABLE: books_silver

CSV rows: 190
CSV rows loaded : 190
MySQL rows : 190
STATUS : SUCCESS

TABLE: customers_silver

CSV rows: 2766
Warning - MySQL columns not present in CSV:
['first_name', 'last_name']
CSV rows loaded : 2766
MySQL rows : 2766
STATUS : SUCCESS

TABLE: orders_silver

CSV rows: 27354
CSV rows loaded : 27354
MySQL rows : 27354
STATUS : SUCCESS

TABLE: loans_silver

CSV rows: 9402
CSV rows loaded : 9402
MySQL rows : 9402
STATUS : SUCCESS

TABLE: reviews_silver

CSV rows: 5454
CSV rows loaded : 5454
MySQL rows : 5454
STATUS : SUCCESS

TABLE: silver_rejected_rows

CSV rejected rows : 1395
MySQL rows : 1395
STATUS : SUCCESS

=====

FINAL MYSQL SILVER VALIDATION

=====

books_silver	: 190
customers_silver	: 2766
orders_silver	: 27354
loans_silver	: 9402
reviews_silver	: 5454
silver_rejected_rows	: 1395

MySQL connection closed.

=====

SILVER MYSQL SYNC COMPLETED

=====

Q7. Implement Gold Layer Business KPIs

Explanation

The Gold layer contains business-ready analytical outputs generated from the validated Silver data.

The following Silver tables were loaded into MySQL:

- books_silver
- customers_silver
- orders_silver
- loans_silver
- reviews_silver

Five business KPIs were created:

- Monthly Revenue Growth
- Customer Retention Rate
- Book Sell-Through Rate
- Library Return Compliance
- Review Coverage Rate

Screenshot

Q8. Calculate Monthly Revenue Growth

Explanation

Monthly revenue growth was calculated by comparing current monthly revenue with the previous month's revenue.

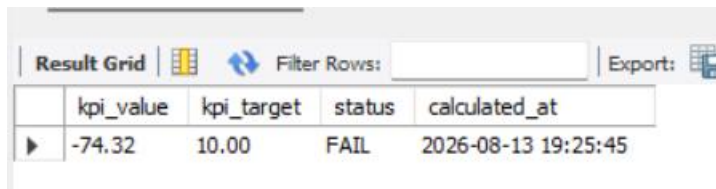
Actual Value: -74.32%

Target Value: 10.00%

Status: FAIL

The KPI did not meet the defined business target. The negative growth value indicates that the current dataset does not satisfy the required monthly revenue growth target.

Screenshot



The screenshot shows a 'Result Grid' interface with a toolbar containing 'Filter Rows' and 'Export' buttons. Below the toolbar is a table with the following data:

	kpi_value	kpi_target	status	calculated_at
▶	-74.32	10.00	FAIL	2026-08-13 19:25:45

Q9. Calculate Customer Retention Rate

Explanation

Customer retention measures the percentage of customers who continued to interact with the library during subsequent periods.

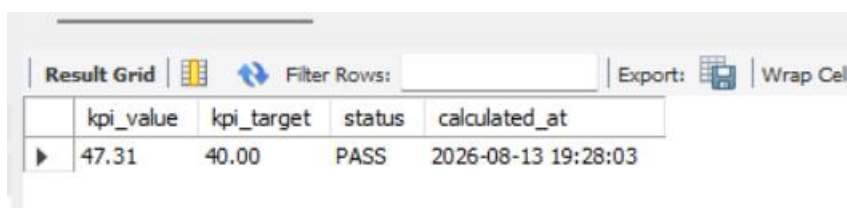
Actual Value: 47.31%

Target Value: 40.00%

Status: PASS

The KPI exceeded the defined business target.

Screenshot



The screenshot shows a 'Result Grid' interface with a toolbar containing 'Filter Rows', 'Export', and 'Wrap Cell' buttons. Below the toolbar is a table with the following data:

	kpi_value	kpi_target	status	calculated_at
▶	47.31	40.00	PASS	2026-08-13 19:28:03

Q10. Calculate Book Sell-Through Rate

Explanation

The Book Sell-Through Rate measures book sales relative to the available stock.

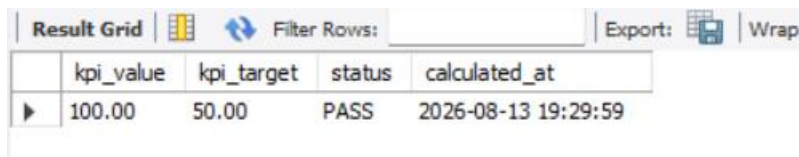
Actual Value: 100.00%

Target Value: 50.00%

Status: PASS

The KPI exceeded the defined business target.

Screenshot



	kpi_value	kpi_target	status	calculated_at
▶	100.00	50.00	PASS	2026-08-13 19:29:59

Q11. Calculate Library Return Compliance

Explanation

Library Return Compliance measures the percentage of loans that were returned within the expected due date.

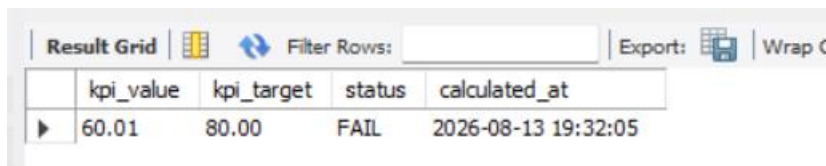
Actual Value: 60.01%

Target Value: 80.00%

Status: FAIL

The KPI did not meet the defined target and therefore requires improvement.

Screenshot



	kpi_value	kpi_target	status	calculated_at
▶	60.01	80.00	FAIL	2026-08-13 19:32:05

Q12. Calculate Review Coverage Rate

Explanation

Review Coverage Rate measures the percentage of books that have received at least one customer review.

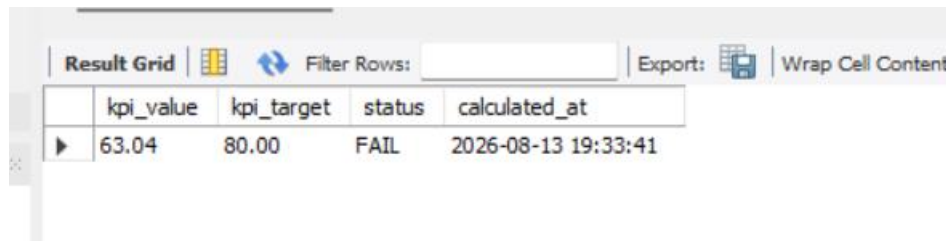
Actual Value: 63.04%

Target Value: 80.00%

Status: PASS

The KPI did not meet the defined business target and requires improvement.

Screenshot



The screenshot shows a software interface with a 'Result Grid' tab. The grid contains one row of data with the following values: kpi_value: 63.04, kpi_target: 80.00, status: FAIL, and calculated_at: 2026-08-13 19:33:41. The interface also includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' option.

kpi_value	kpi_target	status	calculated_at
63.04	80.00	FAIL	2026-08-13 19:33:41

Q13. Perform Pipeline Health Audit

Explanation

A pipeline health audit was implemented to compare the actual KPI values against predefined target values.

The audit generated PASS or FAIL status for each KPI.

KPI	Actual	Target	Status
Monthly Revenue Growth	-74.32%	10.00%	FAIL
Customer Retention Rate	47.31%	40.00%	PASS
Book Sell-Through Rate	100.00%	50.00%	PASS
Library Return Compliance	60.01%	80.00%	FAIL
Review Coverage Rate	63.04%	80.00%	FAIL

Total KPIs: 5

Passed KPIs: 2

Failed KPIs: 3

Screenshot

Result Grid	Filter Rows:	Export:	Wrap Cell
kpi_name	kpi_value	kpi_target	status
Revenue Growth	-74.32	10.00	FAIL
Customer Retention Rate	47.31	40.00	PASS
Book Sell-Through Rate	100.00	50.00	PASS
Library Return Compliance	60.01	80.00	FAIL
Review Coverage	63.04	80.00	FAIL

Q14. Validate Overall Pipeline Health

Explanation

The overall pipeline health was calculated from the individual KPI results.

Total KPIs: 5

Passed KPIs: 2

Failed KPIs: 3

Overall Status: FAIL

The overall status is FAIL because three KPIs did not meet their required targets:

- Monthly Revenue Growth – -74.32% against a target of 10.00%
- Library Return Compliance – 60.01% against a target of 80.00%
- Review Coverage Rate – 63.04% against a target of 80.00%

The audit successfully identifies the KPI that requires attention.

Screenshot

Result Grid	Filter Rows:	Export:	Wrap
kpi_name	actual_value	target_value	status
Monthly Revenue Growth	-74.32	10.00	FAIL
Customer Retention Rate	47.31	40.00	PASS
Book Sell-Through Rate	100.00	50.00	PASS
Library Return Compliance	60.01	80.00	FAIL
Review Coverage Rate	63.04	80.00	FAIL

Q15. Perform Final Pipeline Audit

Explanation

The final pipeline audit provides a consolidated view of the Bronze, Silver, and Gold layers.

The audit validates:

- Bronze ingestion
- Silver rejected records
- Silver row counts
- Gold KPI values
- KPI targets
- PASS/FAIL status

The final audit contains **12 validation checks**.

The final Silver validation confirmed:

- books_silver – 190
- customers_silver – 2,766
- orders_silver – 27,354
- loans_silver – 9,402
- reviews_silver – 5,454
- silver_rejected_rows – 1,395

Screenshot

Result Grid					
		Filter Rows:	Export:  Wrap Cell Content		
	layer	check_name	actual_value	target_value	status
▶	BRONZE	Raw source ingestion	0.00	0.00	PASS
	SILVER	Rejected rows	1395.00	0.00	INFO
	SILVER	Books silver rows	190.00	190.00	PASS
	SILVER	Customers silver rows	2766.00	2800.00	PASS
	SILVER	Orders silver rows	27354.00	28336.00	PASS
	SILVER	Loans silver rows	9402.00	9614.00	PASS
	SILVER	Reviews silver rows	5454.00	5621.00	PASS
	GOLD	Monthly Revenue Growth	-74.32	10.00	FAIL
	GOLD	Customer Retention Rate	47.31	40.00	PASS
	GOLD	Book Sell-Through Rate	100.00	50.00	PASS
	GOLD	Library Return Compliance	60.01	80.00	FAIL
	GOLD	Review Coverage Rate	63.04	80.00	FAIL

Output Summary

Dataset / Metric	Result
Source Records	46,561
Bronze Incremental New Rows	0
Silver Valid Records	45,851
Rejected Records	1395
Gold KPIs	5
Passed KPIs	2
Failed KPIs	3
Overall KPI Status	FAIL

Silver Dataset Rows

Dataset	Rows
Books	190
Customers	2,766
Orders	27,354
Loans	9,402
Reviews	5,454

MySQL Silver Validation

The Silver datasets were successfully synchronized with MySQL.

Table	MySQL Rows Status	
books_silver	190	SUCCESS
customers_silver	2,766	SUCCESS

Table	MySQL Rows	Status
orders_silver	27,354	SUCCESS
loans_silver	9,402	SUCCESS
reviews_silver	5,454	SUCCESS
silver_rejected_rows	1,395	SUCCESS

The final MySQL connection was closed successfully after validation.

Conclusion

This capstone project successfully demonstrated the development of an end-to-end Data Engineering pipeline using the Medallion Architecture.

The source CityReads datasets were loaded into MySQL and processed through the Bronze, Silver, and Gold layers.

The Bronze layer implemented incremental ingestion and watermark-based processing. The Silver layer cleaned and validated the data while separating invalid records into a rejected-record dataset. The Gold layer generated five business KPIs using SQL-based analytical views.

Finally, a pipeline health audit was implemented to compare the KPI results against predefined targets.

The final validation showed that four out of five KPIs passed their targets, while Library Return Compliance failed to meet the required 80% target.

The project provided practical experience in incremental data processing, data quality validation, SQL analytics, business KPI generation, and pipeline monitoring.

References

- MySQL Documentation
- Python Documentation
- Pandas Documentation
- SQL Documentation
- Celebal Technologies Data Engineering Internship Material
- GitHub Documentation